Wilson-Cowan-type neural activity models

In Wilson-Cowan-type models, the activity of a neural population is expressed in terms of synaptic activity u, as follows

$$\tau \dot{u} = -u + tv + h$$

where t is the proportionality constant between firing rate v and synaptic activity u. (Here I am assuming just one population so $t > 0$ represents intrinsic recurrent excitation.) The parameter h represents synaptic activity due to extrinsic sources.

Often the following linear relationship between synaptic activity u and the resulting firing rate v is assumed,

$$v = gu + v_0 \, ,$$

where $g > 0$ and v_0 is the spontaneous firing rate (firing rate in the absence of synaptic activity). Differentiating this relationship gives $\dot{v} = g\dot{u}$ so

$$\frac{\tau}{g}\dot{v} = -\frac{v - v_0}{g} + tv + h$$

$$\tau \dot{v} = -v + \underbrace{gt}_{w} v + \underbrace{gh + v_0}_{f}$$